CNAG MVP: Multi variant pipeline

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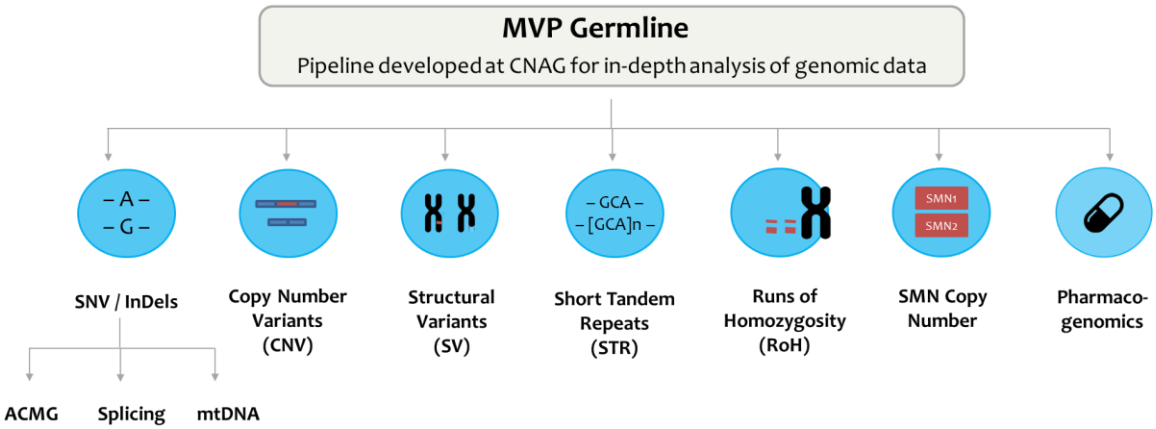
This document describes the different results generated by the CNAG multi-variant pipeline (MVP) and provides instructive insights for the filtering and interpretation of variants. All files are available on the CNAG's ftp server and can be downloaded using ftp clients such as lftp, WinSCP and Filezilla. You will have received an email with credentials to securely download your data.

Results are generated in a research context and must be validated for diagnostic purposes.



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1. SNV-InDels

All SNV-InDels are available for filtering and interpretation in the GPAP platform: https://portal.gpap.cnag.eu/nextgpap_38/#/

In addition, annotated and pre-filtered data with candidate gene lists is provided as complementary results. SNVs and Indels are called with GATK-HaplotypeCaller[1], annotated with SNPEff[2], GREEN-VARAN[3] and SpliceAI[4].

Folder Name: SNV-InDels

2. Structural variants

All structural variants are available for filtering and interpretation in the GPAP platform: https://portal.gpap.cnag.eu/nextgpap_38/#/ .
In addition, pre-filtered data with candidate gene lists is provided as complementary results.

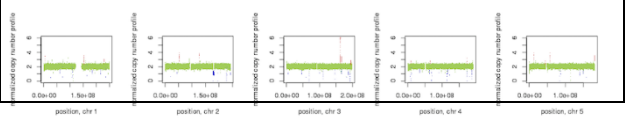
a. Copy number variants: CNVs

CNVs are called with Control-FREEC[5]with 3 sliding window sizes (5000bp, 20000bp and 50000bp) comparing to a baseline generated from 50 individuals sequenced with the same conditions. Results are annotated with AnnotSV[6]. Smaller window sizes lead to the detection of a higher number of CNVs.

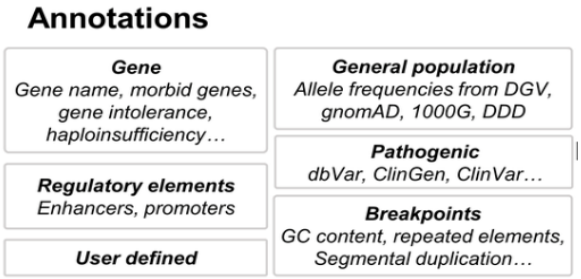
Please be aware that CNV tools are based on sequence read coverage which is subject to technical limitations when using short read sequencing and may yield a significant number of false positives. It is highly advisable to visually validate variants by examining the alignments (refer to section 9: Genomic Alignments Visualization) and validate strong candidates by an orthogonal method.

Another set of results is provided using ClinSV[7], based on CNVnator[8] and Lumpy[9] as complementary callers.

Folder name: Structural_variants

Analysis	File name	Filtering	Suggested filtering and interpretation
CNVs – large deletions and duplications	BARCODE.ControlFreeC_<windowsize>.CNV.png	Overview of dosage gain and loss at the chromosome level	The pattern observed should be very homogeneous as shown in this example: 

1 https://gatk.broadinstitute.org/hc/en-us/articles/360037225632-HaplotypeCaller
2 https://pcingola.github.io/SnpEff/
3 https://green-varan.readthedocs.io/en/latest/
4 https://github.com/Illumina/SpliceAI
5 http://boevalab.inf.ethz.ch/FREEC/
6 https://lbgi.fr/AnnotSV/
7 https://genomemedicine.biomedcentral.com/articles/10.1186/s13073-021-00841-x
8 https://github.com/abyzovlab/CNVnator
9 https://github.com/arq5x/lumpy-sv

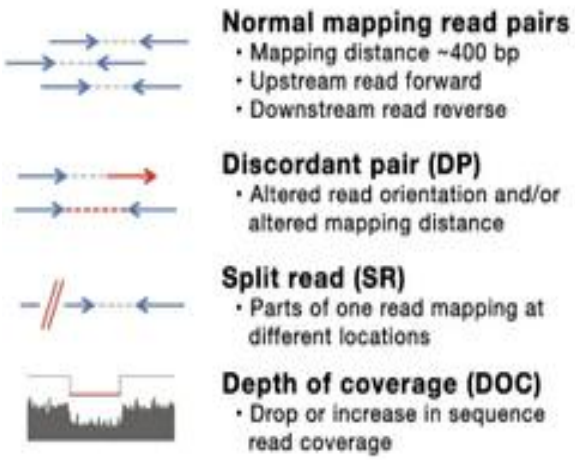
			Dosage loss is shown in blue and gain in red. Poor sample quality can result in non-uniform patterns, contributing to an increased occurrence of false positive CNVs.
CNVs- All and candidate gene lists	BARCODE.ControlFreeC_<windowsize.CNV.annot.tsv	CNVs detected with ControlFREEC[10] with the sliding window size indicated in the file name. Results are annotated with AnnotSV[11]	 Detailed information on AnnotSV annotations can be found here[12] . Each CNV is reported in two ways: "full" = complete CNV including 0-n genes and "split" = CNV divided by gene(s). For example, if a CNV encompass 3 genes, results will show 4 rows for the same CNV: 1 full, and 3 split (one row per gene). Start by using the full mode and filter by the AnnotSV ranking score 4 (likely pathogenic) and 5 (pathogenic): AnnotSV Score / ACMG class ≥ 0.99 — Class 1 (benign) 0.90 to 0.98 — Class 2 (likely benign) 0.89 to -0.89 — Class 3 (VUS) −0.90 to −0.98 — Class 4 (likely pathogenic) ≤ -0.99 — Class 5 (pathogenic) Discard CNVs completely contained within an established benign CNV region (columns from AX to BI, considering the type of variant). Check for concordant phenotypes in columns OMIM_phenotype (column CS), GenCC_disease (column CN) and inheritance (column CT) OR cross with corresponding gene list. Check reported pathogenic duplication, deletion or insertion (columns AJ to AW). For interpretation consider: OMIM morbid (CU)(if yes the gene is considered essential) and the haploinsufficiency and triple sensitivity status (columns CA and CB). For candidate variants, we recommend checking

[10] http://boevalab.inf.ethz.ch/FREEC/
[11] https://lbgi.fr/AnnotSV/
[12] https://lbgi.fr/AnnotSV/Documentation/README.AnnotSV_latest.pdf.

			the corresponding gene / variant in ClinGen for dosage sensitivity [13]and other known pathogenic variants.
CNVs-Additional output: ClinSV	BARCODE.ClinSV_RARE_PASS_GENE.CNV.annot.tsv	CNVs detected with ClinSV and annotated with the candidate gene list (if provided)	Use this additional output if no causative variants are found in previous results. Start filtering by coding regions and select exon, CDS or start codon in the "GFEAT" column. To filter out variants with poor quality and reduce the number of possible false positives, use Column SU (Sum of discordant pairs and split reads supporting the variant) and exclude if = 0. For clinical interpretation use column "PHEN" to identify phenotypes compatible with you patient. For interpretation consider column "PHEN" to identify phenotypes compatible with you patient. For candidate variants, we recommend checking the corresponding gene / variant in ClinGen for dosage sensitivity [14]and other known pathogenic variants.

b. Other structural variants

Structural Variants are called with Manta[15] and annotated with AnnotSV [16].
SV tools are based on discordant read pairs, split reads and depth of coverage as shown below:



Please be aware that these approaches are subject to technical limitations when using short read sequencing and may yield a significant number of false positives. It is highly advisable to visually validate variants by examining the alignments (refer to section 9: Genomic Alignments Visualization) and validate strong candidates by an orthogonal method.

[13] https://search.clinicalgenome.org/kb/gene-dosage?page=1&size=25&search=
[14] https://search.clinicalgenome.org/kb/gene-dosage?page=1&size=25&search=
[15] https://github.com/Illumina/manta
[16] https://lbgi.fr/AnnotSV/

Folder name: Structural_variants

Analysis	File name	Filtering	Suggested filtering and interpretation
Structural variants	BARCODE. manta.annot.tag.tsv	SVs detected with Manta and annotated with AnnotSV	**Annotations** ***Gene*** *Gene name, morbid genes, gene intolerance, haploinsufficiency…* ***General population*** *Allele frequencies from DGV, gnomAD, 1000G, DDD* ***Pathogenic*** *dbVar, ClinGen, ClinVar…* ***Regulatory elements*** *Enhancers, promoters* ***Breakpoints*** *GC content, repeated elements, Segmental duplication…* ***User defined*** Detailed information on AnnotSV annotations can be found here[17] Each SV is reported in two ways: "full" = complete SV including 0-n genes and "split" = SV divided by gene(s). For example, if a SV encompasses 3 genes results will show 4 rows for the same SV: 1 full, and 3 split (one row per gene). Start by using the full mode and filter by the AnnotSV ranking score 4 (likely pathogenic) and 5 (pathogenic)(note that it does not include classification for inversions): **AnnotSV Score** / **ACMG class** ≥ 0.99 — Class 1 (benign) 0.90 to 0.98 — Class 2 (likely benign) 0.89 to -0.89 — Class 3 (VUS) −0.90 to −0.98 — Class 4 (likely pathogenic) ≤ -0.99 — Class 5 (pathogenic) Discard SVs completely contained within an established benign SV region (columns from AX to BI, considering the type of variant). Check for concordant phenotypes in columns OMIM_phenotype (column CS), GenCC_disease (column CN) and inheritance (column CT) OR cross with corresponding gene list. Check reported pathogenic duplication, deletion or insertion (columns AJ to AW). For interpretation consider: OMIM morbid (CU)(if yes the gene is considered essential) and the haploinsufficiency and triple sensitivity status (columns CA and CB). For candidate variants, we recommend checking the corresponding gene / variant in ClinGen for dosage sensitivity [18]and other known pathogenic variants.

[17] https://lbgi.fr/AnnotSV/Documentation/README.AnnotSV_latest.pdf.
[18] https://search.clinicalgenome.org/kb/gene-dosage?page=1&size=25&search=

3. mtDNA

Mitochondrial SNVs and InDels are called with GATK for mitochondria[19] based on mutect2[20]. Results are annotated with gnomAD, MitoMAP[21] and MitoTIP[22] for disease-gene interpretation.

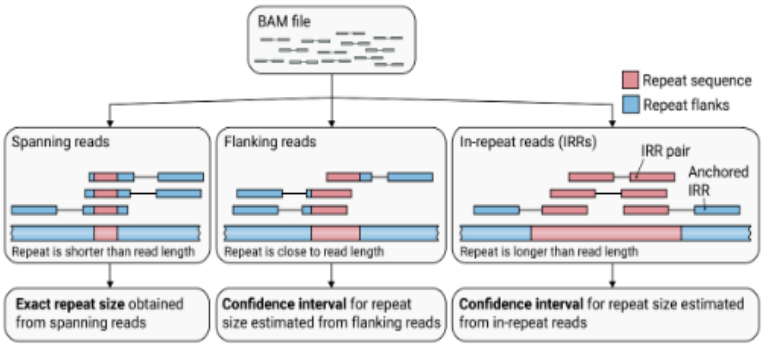
Folder name: mtDNA

Analysis	File name	Filtering	Suggested filtering and interpretation
mtDNA	BARCODE.MT_GATK.annot.xlsx	SNV-Indels in the mitochondrial chromosome with heteroplasmy >1%	Start filtering out low quality variants by selecting "PASS" in the QC PASS column. Use Mitomap reported status annotations to filter by variants that have already been "reported" or "confirmed" as pathogenic in this database. Consider filtering out variant with low heteroplasmy (<10%) as callers are poorly sensitive below this range and return many false positives (>0.01 in the barcode_AF column). For interpretation consider reviewing annotations from Mitomap[23] and checking phenotypic descriptions in MitoPhen[24] and/or hmtvar[25]. If the data from the mother is available – check the candidate variant status and if the heteroplasmy/homoplasmy levels are compatible with the segregation of the disease.

4. Short tandem repeats: STRs

Known disease-causing repeat expansions are called with ExpansionHunter [26]. This tool estimates the size of 36 known pathogenic STRs from 29 genes by assessing spanning, flanking and in-repeat reads as shown here:

19 https://gatk.broadinstitute.org/hc/en-us/articles/4403870837275-Mitochondrial-short-variant-discovery-SNVs-Indels-
20 https://gatk.broadinstitute.org/hc/en-us/articles/360037593851-Mutect2
21 https://www.mitomap.org/MITOMAP
22 https://www.mitomap.org/MITOMAP/MitoTipInfo
23 https://www.mitomap.org/allelesearch.html
24 https://mitophen.org/
25 https://www.hmtvar.uniba.it/
26 Dolzhenko et al 2017 and 2019; https://github.com/Illumina/ExpansionHunter/blob/master/docs/06_OutputVcfFiles.md



Folder name: ShortTandemRepeats

Analysis	File name	Filtering	Suggested filtering and interpretation
STRs – known disease causing	BARCODE.Stripy_STRs	Repeat expansions known to be disease causing	Check genes and alignments of results out of range – highlighted in red.

Additionally, the ExpansionHunter long catalogue provides an extended set of repeat expansions across the genome, beyond the limited set of clinically relevant loci included in the short catalogue, enabling broader discovery and analysis of novel or less-characterized repeats.

5. Pseudogenes

a. GBA

Folder name: Pseudogeness -> GBA

Analysis	File name	Filtering	Suggested filtering and interpretation
GBA	BARCODE.GBA.tsv	GBA carriers or affected individuals	Description of the results annotations:

Fields in tsv	Explanation
Sample	Sample name
is_biallelic(GBAP1-like_variant_exon9-11)	Whether the sample is called as biallelic for GBAP1-like variants in exon9-11
is_carrier(GBAP1-like_variant_exon9-11)	Whether the sample is called as a carrier for GBAP1-like variants in exon9-11
CN(GBA+GBAP1)	Total copy number of GBA+GBAP1
deletion_breakpoint_in_GBA	Whether the deletion breakpoint is in GBA gene if a deletion exists
GBAP1-like_variant_exon9-11	GBAP1-like variants called in exon9-11, two alleles separated by /
other_unphased_variants	Other variants called (non-GBAP1-like variants or variants outside of exon9-11)

			Filter column is_biallelic and/or is_carrier to identify possible causative variants.

b. SMN1

This result considers the copy number of full-length *SMN1*, full-length *SMN2*, as well as SMN2Δ7–8 (SMN2 with a deletion of Exon7-8) to test for Spinal Muscular Atrophy affected individuals and carriers. Variants in *SMN1* and the *SMN2* pseudogene are called with SMNCopyNumberCaller[27] .

Folder name: Pseudogenes

Analysis	File name	Filtering	Suggested filtering and interpretation
SMA	BARCODE_SMA.tsv	SMA carriers or affected individuals	Description of the results annotations: Fields in tsv: Explanation Sample: Sample name isSMA: Whether sample is called as SMA (zero copy of SMN1) isCarrier: Whether sample is called as SMA carrier (one copy of SMN1) SMN1_CN: Copy number of SMN1 (None means no-call) SMN2_CN: Copy number of SMN2 SMN2delta7-8_CN: Copy number of SMN2Δ7–8 (deletion of Exon7-8) Total_CN_raw: Raw normalized depth of total SMN Full_length_CN_raw: Raw normalized depth of full-length SMN g.27134T>G_CN: CN of g.27134T>G, SNP associated with 2+0 silent carrier SMN1_CN_raw: Raw SMN1 CN values at SNP sites that differ btn SMN1/SMN2 Filter column isSMA and/or isCarrier to identify possible causative variants.

6. Homozygosity

Runs of Homozygosity (ROH) are computed with Plink[28], which allow to estimate possible familiar consanguinity or uniparental disomy. For consanguinity predictions, we use the sum of the size of all the runs >1,5Mb within autosomal chromosomes and we compare them to the thresholds established in Matalonga et al 2020[29].

Folder name: Homozygosity

Analysis	File name	Filtering	Suggested filtering and interpretation
ROH	BARCODE.hom	Sum of all homozygous regions in	Consider Uniparental disomy upon identification of a single segment of ROH >10 Mb or, two or more segments >5 Mb each involving a single

[27] https://github.com/Illumina/SMNCopyNumberCaller
[28] https://www.cog-genomics.org/plink/
[29] Matalonga L, Laurie S, Papakonstantinou A, et al. Improved Diagnosis of Rare Disease Patients through Systematic Detection of Runs of Homozygosity [published correction appears in J Mol Diagn. 2023 May;25(5):311]. *J Mol Diagn*. 2020;22(9):1205-1215. doi:10.1016/j.jmoldx.2020.06.008

<table>
<tr>
<td></td>
<td></td>
<td>autosomal chromosomes for consanguinity. And identification of large ROH for uniparental disomy.</td>
<td>chromosome (Sahoo et al 2014[30]).Identify possible consanguinity by adding up the size of all regions (exclude sexual chromosomes) and compare with established thresholds[33]:
<table>
<tr><th>ROH interval, Mb</th><th>Experimental offspring consanguinity classification</th></tr>
<tr><td>>123</td><td>Consanguineous</td></tr>
<tr><td>79—123</td><td>Probably consanguineous</td></tr>
<tr><td>22—79</td><td>Uncertain</td></tr>
<tr><td><22</td><td>Nonconsanguineous</td></tr>
</table>
</td>
</tr>
</table>

7. Pharmacogenomics

a. Pharmacogenes

Results from the genotyping of a total of 87 pharmacogenes based on PyPGX[31]. Guidelines on how to interpret results can be found here: https://pypgx.readthedocs.io/en/latest/readme.html#results-interpretation
Output: BARCODE.PyPGx.tsv

b. CYP2D6

Genotyping of the CYP2D6 gene based on Cyrius from Illumina: https://github.com/Illumina/Cyrius
Output: BARCODE.CYP2D6.tsv

8. Immunology

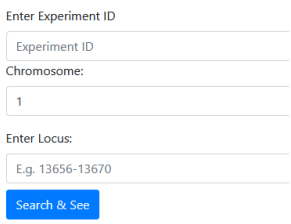
a. MHC: Major histocompatibility complex

i. HLA and KIR

Results from the genotyping of HLA and KIR polymorphic genes using https://github.com/mourisl/T1K and https://www.genome.med.kyoto-u.ac.jp/HLA-HD/
Output 1: BARCODE.HLA.txt
Output 2: BARCODE.MHC.txt

9. Genomic alignments visualisation

An embedded version of the IGV Browser is available in CNAG GPAP at https://portal.gpap.cnag.eu/igvgpap/ . Enter your sample ID and the position or region you would like to access to visualise the corresponding genomic alignments.



[30] https://www.nature.com/articles/ejhg201453
[31] https://pypgx.readthedocs.io/en/latest/readme.html

Please note that the IGV version from the GPAP does not have all the regular IGV functionalities. For in depth genomic visualization you can download the BAM files and open them with a local IGV instance available here.